

JOSEPH SCIOTTO

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Education

Columbia University

Sep. 2023 – May 2027

Bachelor of Science in Mechanical Engineering; Minor in Applied Mathematics; GPA: 3.48/4.00

New York, NY

Active Secret Clearance holder, granted 3/18/24.

Relevant Coursework: Turbomachinery, Mechanics and Thermodynamics of Propulsion, Aerodynamics, Materials & Manufacturing, Fluid Mechanics, Thermodynamics, Analysis and Optimization

Experience

STORC

February 2026 – Present

Electromechanical Engineering Intern

Remote

- Designed and revised a vehicle-mounted electronics enclosure for prototyping and future injection molding
- Coordinated with an electrical engineer to integrate PCB layout constraints, component clearances, and required placement radii into enclosure geometry
- Evaluated aerodynamics, packaging, cost, and manufacturability constraints for 3D-printed prototypes and injection-molded production designs
- Created technical reports for cross-functional teams, including Venture, Finance, and Public Affairs

Columbia University

January 2026 – Present

Course Assistant

New York, NY

- Assisted professor for college-level Drafting and SolidWorks course across 56 students in two sections
- Graded assignments, homework, labs, and exams while providing detailed and constructive technical feedback
- Held in-person office hours to support students with CAD modeling, drafting standards, and course assignments

Hayduk Engineering LLC

June 2024 – August 2024

Wastewater Engineer Intern

Ronkonkoma, NY

- Modeled a full wastewater treatment facility in Revit, coordinating current and planned construction diagrams with expansion capacity for future design review
- Produced process flow diagrams for neighborhood-wide pre-treatment facilities; reduced review iterations by senior engineers by 50%
- Drafted elevator service electrical diagrams in AutoCAD, aligning with NEC constraints and site conditions
- Collaborated with civil team in Civil 3D to adjust site layout and perform runoff calculations for different material grades

Projects

Bugatti Type 51 Model | *SolidWorks, 3D Printing, 3-Axis Milling*

March 2025

- Developed a 3D assembly in SolidWorks of the 1931 racecar, applying geometric modeling for silicone replication
- 3D printed a body mold for silicone casting, accounting for sprue placement and pour hole geometry
- Used 3-axis CNC milling to create an aluminum negative of the model wheels for later plastic injection molding

Plus-Minus Playground Toy | *SolidWorks, FEA*

April 2025

- Developed a playground toy concept to teach young children addition and subtraction without traditional counting methods
- Modeled a see-saw mechanism in SolidWorks using sliders along a number line and an arced ruler to display results
- Used SolidWorks FEA to simulate a child stepping on the toy and impact from a significant drop, achieving a FoS of 1.7

Low-Cost Poiseuille Viscometer | *3D Printing, Arduino, Python*

December 2025

- Designed a low-cost viscometer for medical fluid analysis using syringes and needles to dispense liquid onto a load cell
- Programmed an Arduino to collect mass vs. time data and export measurements to a spreadsheet for analysis
- Used Pandas and NumPy to numerically fit the first-order ODE, achieving 0.62% error in the best trial

Technical Skills

Design: SolidWorks, Fusion 360, Revit, Civil3D, engineering drawings, GD&T

Programming: Python, MATLAB, HTML/CSS, JavaScript, C++, Golang

Manufacturing: 3-axis CNC Milling, 3D Printing, Laser Cutting, Soldering, CNC/Manual Lathe

Simulation/Analysis: SolidWorks FEA, ANSYS CFD, Simulink

Leadership / Extracurricular

ROTC

September 2023 – Present

Cadet

Fordham University

- Led a company of 60 cadets during training and exercise operations throughout a semester
- Managed cadet accountability, attendance, and gear distribution while leading smaller squad-level groups
- Led simulated tactical exercises as a platoon leader to plan and execute assigned missions